

Breast Cancer Detection: Wisconsin Diagnostic Dataset

ISA 640 – Capstone Project

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Project Recap:

Breast cancer remains one of the most common cancers affecting women worldwide, and early, accurate diagnosis is critical to improving treatment outcomes.

This project uses the **Wisconsin Breast Cancer diagnostic dataset**.

Milestones #2:

This project milestone #2 (EDA & Feature Engineering) covers the following:

1. Handle missing values
2. Address data imbalance
3. Feature engineering / dimensionality reduction
4. Show the impact of preprocessing
5. Summary statistics
6. Visualizations
7. Identify patterns tied to our hypothesis
8. Models trained afterward: Logistic Regression, KNN, SVM, Decision Tree, Random Forest, and Ensembles (Voting/Bagging/Boosting/Stacking).

Note: **Malignant = 1, Benign = 0.**

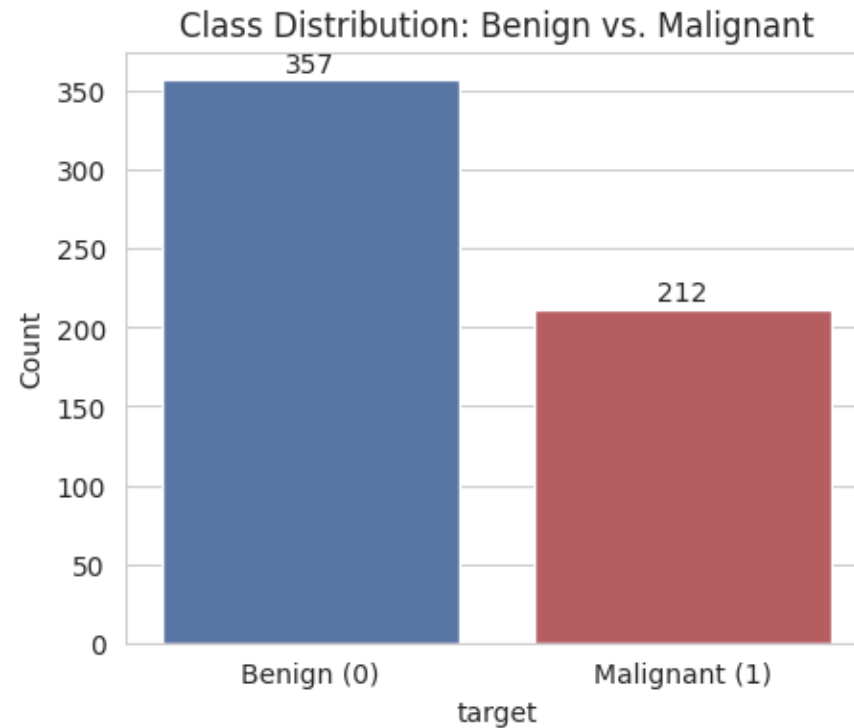
Missing values:

Total missing values in the dataset: 0

...	Total missing values in the dataset: 0	
	missing_count	missing_pct
mean radius	0	0.0
mean texture	0	0.0
mean perimeter	0	0.0
mean area	0	0.0
mean smoothness	0	0.0
mean compactness	0	0.0
mean concavity	0	0.0
mean concave points	0	0.0
mean symmetry	0	0.0
mean fractal dimension	0	0.0
radius error	0	0.0
texture error	0	0.0
perimeter error	0	0.0
area error	0	0.0
smoothness error	0	0.0
compactness error	0	0.0
concavity error	0	0.0
concave points error	0	0.0

Data Imbalance:

Imbalance ratio (benign : malignant) is 1.68 : 1



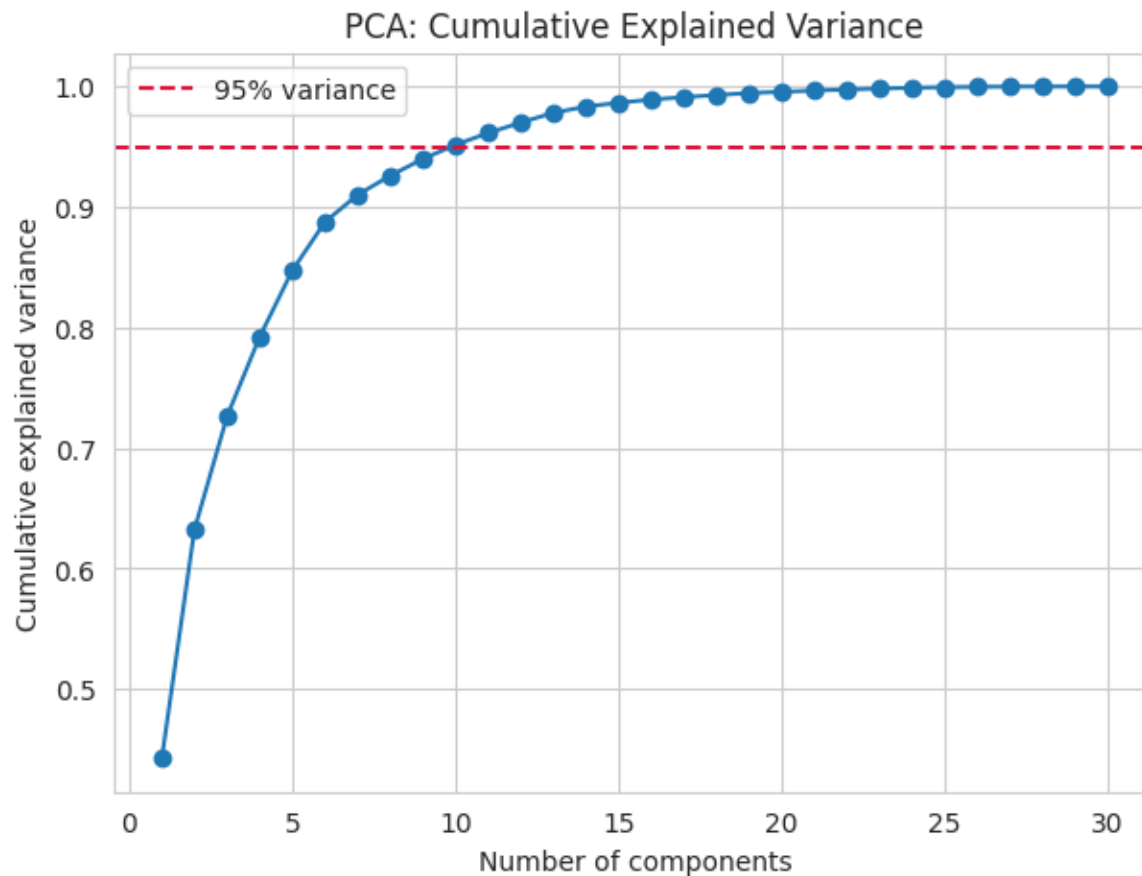
Features Engineering:

- I create a few ratio/interaction features motivated by the biological hypothesis that malignant tumors tend to be **larger, more irregular in shape, and more heterogeneous** than benign ones:

	count	mean	std	min	25%	50%	75%	max
worst_mean_radius_ratio	569.0	1.142040	0.081259	1.0	1.087405	1.120755	1.177966	1.589189
concavity_compactness_ratio	569.0	0.728382	0.403049	0.0	0.438014	0.692245	1.010039	2.569090
radius_concavepoints_interaction	569.0	0.803372	0.849515	0.0	0.238560	0.439215	1.128672	5.111916
worst_mean_texture_ratio	569.0	1.331644	0.130911	1.0	1.236426	1.318646	1.408341	2.067160

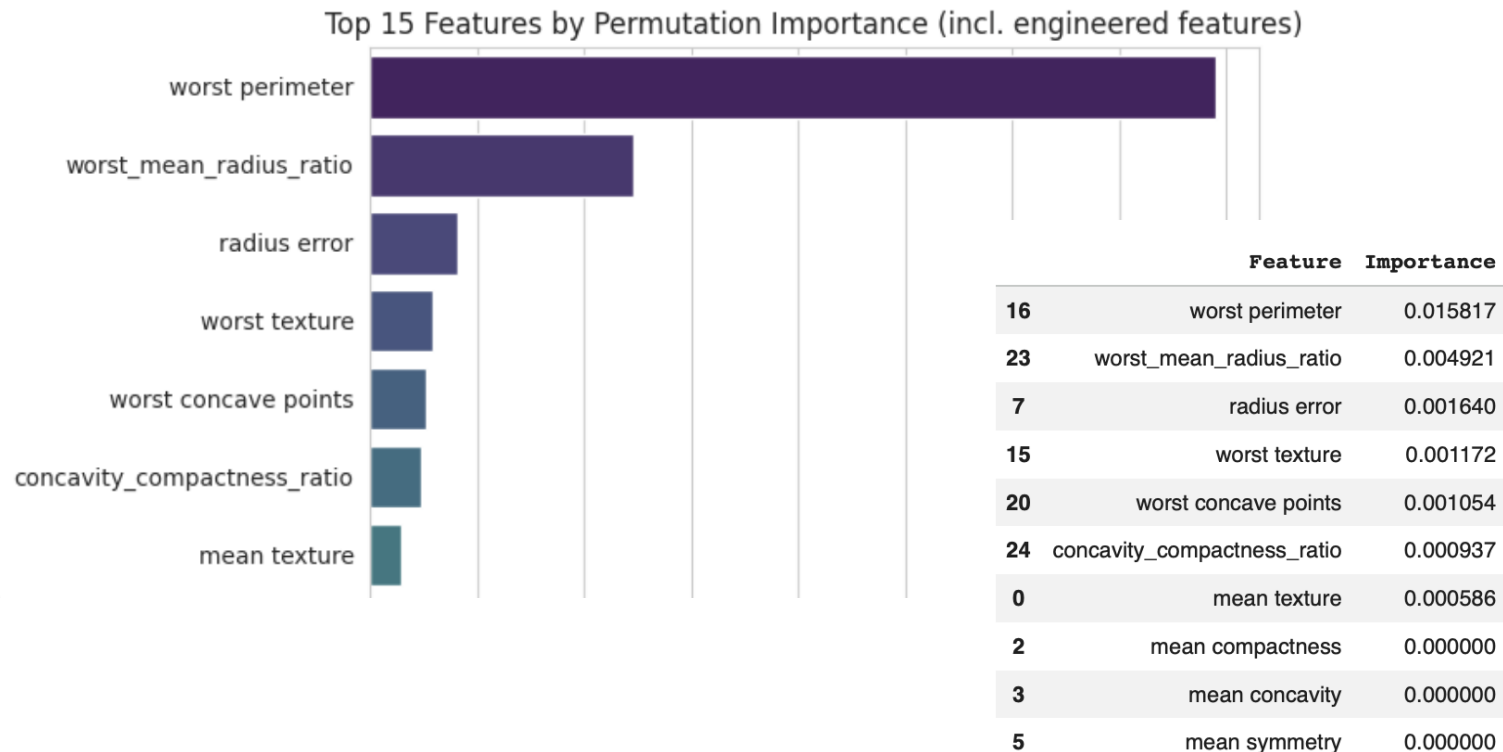
Features Engineering:

- Scaling & PCA

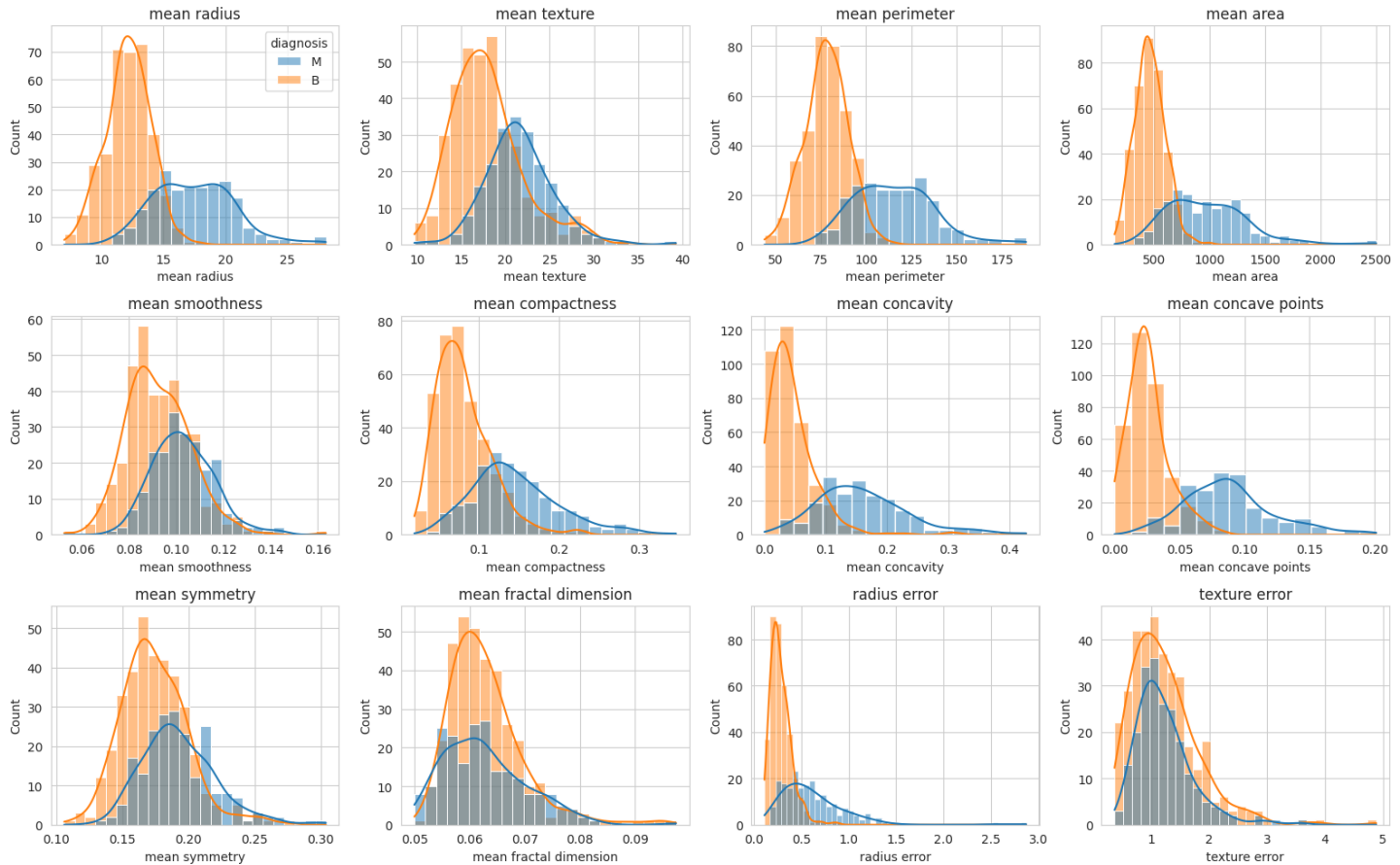


Features Engineering:

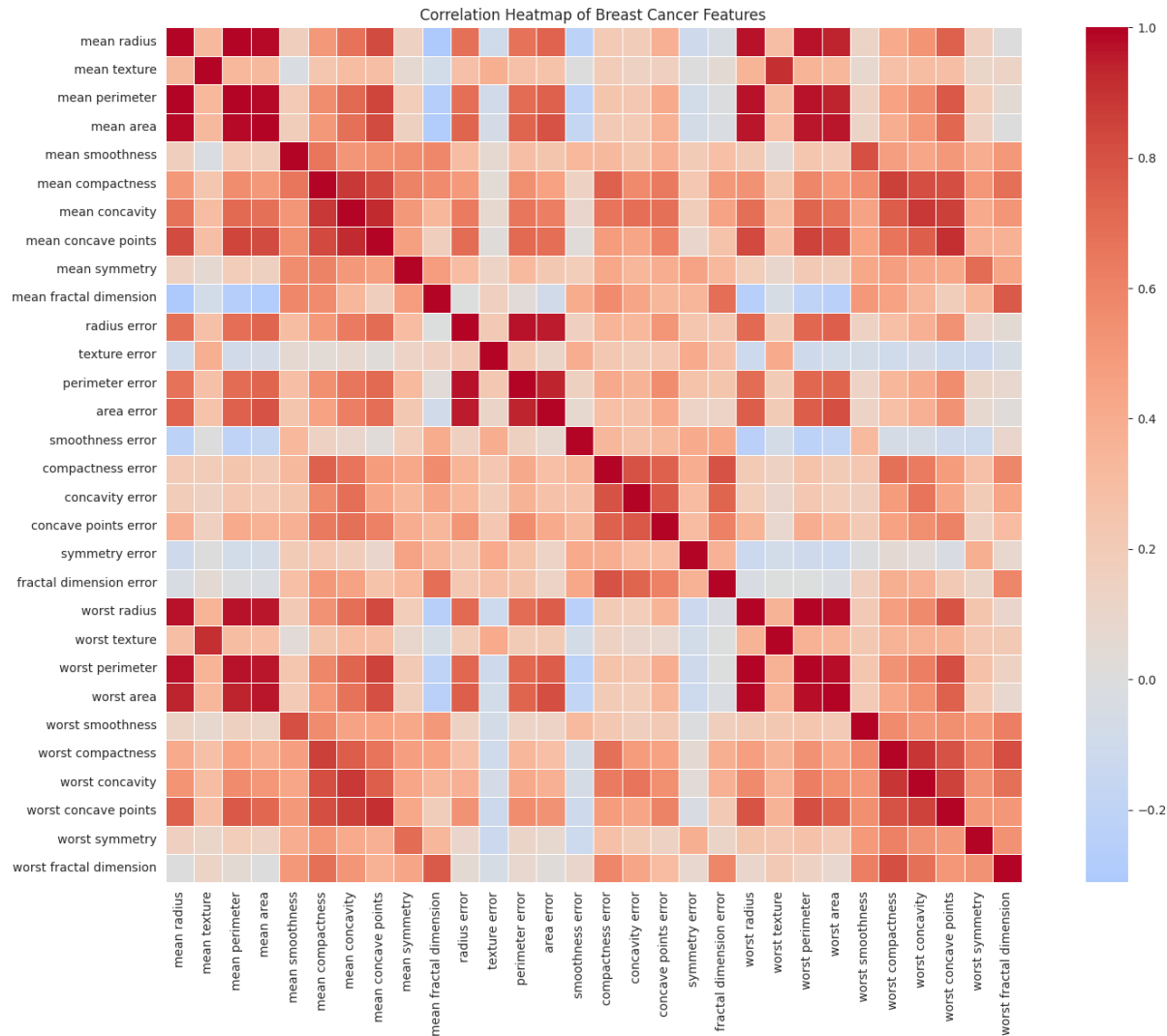
- Features Importance



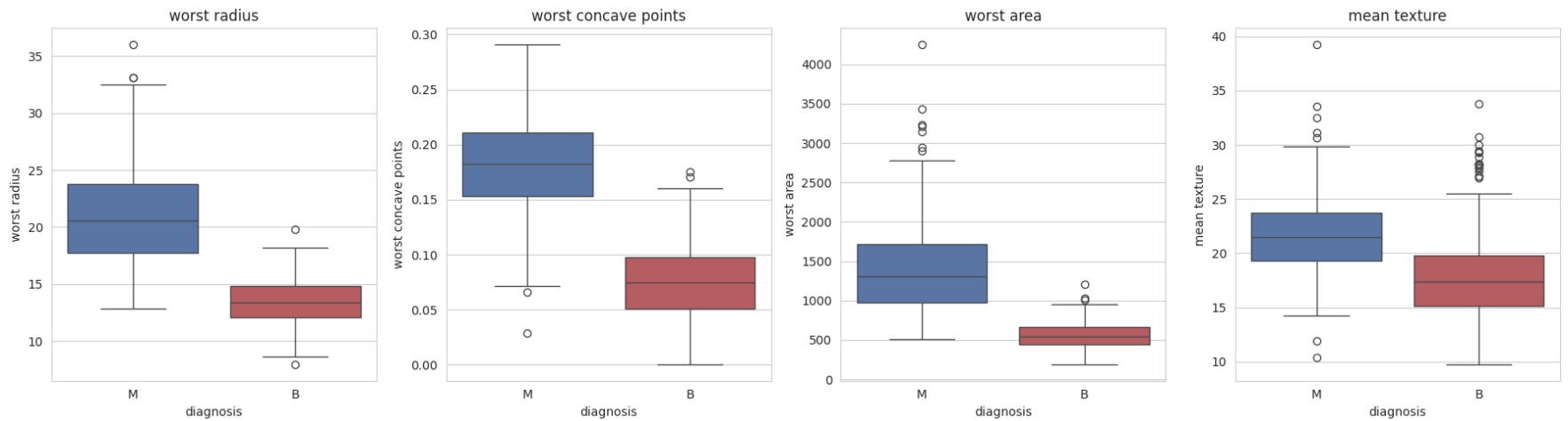
EDA- Features by Class:



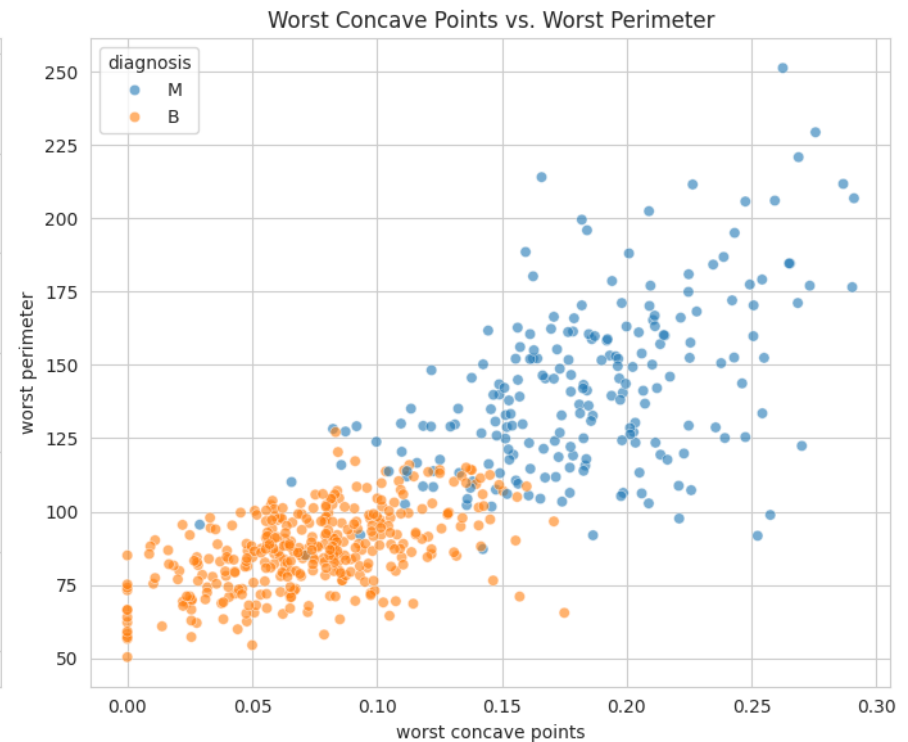
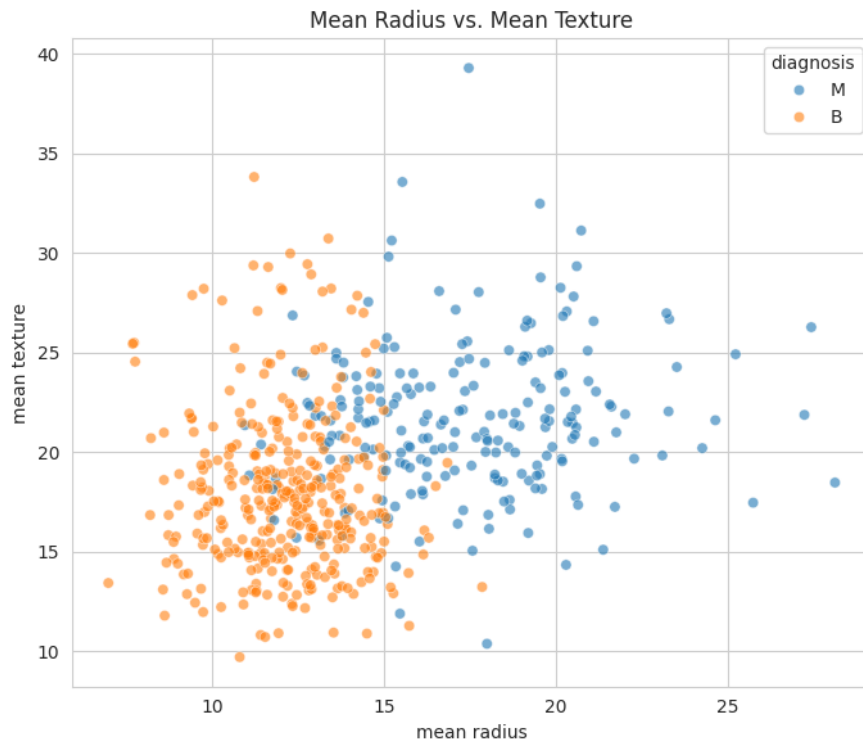
EDA- Correlation Heatmap:



EDA- Key Features by Class:



EDA- Relationship between Variables:



Key Takeaways:

- **Size and shape** features (radius, perimeter, area, concavity, concave points) clearly distinguish malignant from benign tumors.
- Malignant tumors are generally larger and have more irregular shapes across both mean and worst measurements.
- The **worst concave points** vs. worst perimeter scatterplot shows strong class separation, supporting the effectiveness of combining size and shape features.
- The correlation heatmap reveals **high redundancy** among many *mean*, *SE*, and *worst* versions of the same measurements (correlation > 0.95), justifying feature reduction.

Thank you!

Questions?

